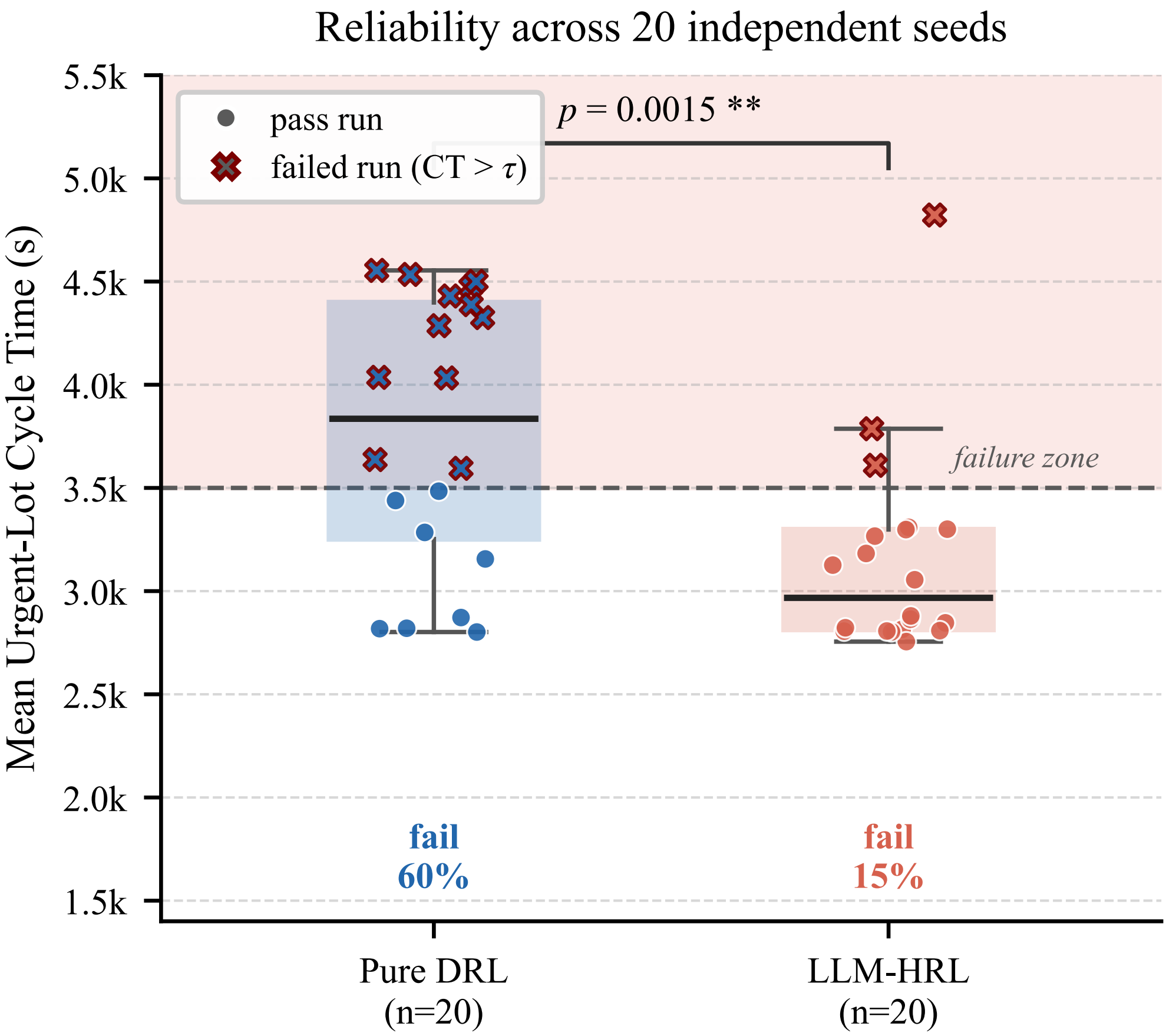
Reliability across 20 independent seeds
pass run
failed run (CT > τ)
p = 0.0015 **
failure zone
Mean Urgent-Lot Cycle Time (s)
5.5k
5.0k
4.5k
4.0k
3.5k
3.0k
2.5k
2.0k
1.5k
fail
60%
fail
15%
Pure DRL
(n=20)
LLM-HRL
(n=20)